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A new reproducing kernel-based nonlinear dimension reduction method for survival data

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Abstract

Based on the theories of sliced inverse regression (SIR) and reproducing kernel Hilbert space (RKHS), a new approach RDSIR (RKHS-based Double SIR) to nonlinear dimension reduction for survival data is proposed. An isometric isomorphism is constructed based on the RKHS property, then the nonlinear function in the RKHS can be represented by the inner product of two elements that reside in the isomorphic feature space. Due to the censorship of survival data, double slicing is used to estimate the weight function to adjust for the censoring bias. The nonlinear sufficient dimension reduction (SDR) subspace is estimated by a generalized eigen-decomposition problem. The asymptotic property of the estimator is established based on the perturbation theory. Finally, the performance of RDSIR is illustrated on simulated and real data. The numerical results show that RDSIR is comparable with the linear SDR method. Most importantly, RDSIR can also effectively extract nonlinearity from survival data.

KEYWORDS

operator, perturbation theory, RKHS, sliced inverse regression, sufficient dimension reduction subspace, survival data

Wenquan Cui and Jianjun Xu contributed to the work equally and should be regarded as co-first authors.

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1 | INTRODUCTION

High-dimensional data are encountered in a wide range of areas including biology, genomics, health sciences, astronomy, economics and machine learning. Variable selection and sufficient dimension reduction (SDR) are two commonly used methods in modeling such data. However, these two methods are based on different assumptions. For variable selection, researchers assume that among all the covariates, only a few are truly related to the response. In recent years, researchers have done much work on variable selection and gained notable achievements including LASSO type (Sun & Zhang, 2012; Tibshirani, 1996; Yuan & Lin, 2006; Zou, 2006), elastic net (Zou & Hastie, 2005), Dantzig selector (Candes & Tao, 2007), SCAD (Fan & Li, 2001), SIS (Fan & Lv, 2008; Fan & Song, 2010), and many others. While in the aspect of SDR, the assumption is that the response may relate to all the covariates but only relates to a few linear combinations of them. The goal of SDR is to recover the space spanned by the coefficient vectors of these linear combinations. There are several commonly used methods in the existing literature: the inverse regression methods (e.g., Li, 1991), the nonparametric methods (e.g., Xia et al., 2002), the semiparametric methods (e.g., Ma & Zhu, 2012) and the optimization methods (e.g., Sheng & Yin, 2016). The inverse regression methods started with sliced inverse regression (SIR) proposed by Li (1991), which considered the following model

$$Y = g(\beta_1^T \mathbf{X}, \dots, \beta_q^T \mathbf{X}, \epsilon), \quad (1)$$

where $\mathbf{X}$ is the p -dimensional regressor vector, Y is the response variable, g is an arbitrary function on $\mathbb{R}^{q+1}$, the random error ϵ is independent of $\mathbf{X}$, and $\beta_1, \dots, \beta_q$ ($q \leq p$) are p -dimensional nonrandom vectors which are so-called SDR directions. The space generated by SDR directions is called SDR subspace and obviously, $\{\beta_1, \dots, \beta_q\} = I_p$ is a trivial case when $p = q$, that is we have p column vectors in the SDR subspace. Let $\mathcal{B}$ be the central subspace which is the intersection of all SDR subspaces. Under mild conditions (Cook, 2009; Yin et al., 2008), the central subspace $\mathcal{B}$ exists and is unique. The goal of SDR under model (1) is to find $\mathcal{B}$ without specifying the unknown function g . To identify the central space, SIR uses the first several significant eigenvectors of the matrix $\text{Cov}[E(\mathbf{X}|Y)]$ to recover $\mathcal{B}$. However, SIR fails to identify some symmetric patterns. Sliced average variance estimation (SAVE, Cook & Weisberg, 1991) was developed to deal with this problem. The regularized SIR (RSIR) proposed by Zhong et al. (2005) is efficient and stable in computation for data with high dimensionality and high collinearity. Wu et al. (2013) discussed some asymptotic properties of RSIR. Some other inverse regression methods are directional regression (DR, Li & Wang, 2007), kernel inverse regression (Zhu et al., 1996), and so on. All these methods are based on certain conditions including the linearity condition (Li, 1991) and the constant variance condition (Cook & Weisberg, 1991). As for nonparametric methods that are conceptually more intuitive while computationally more complicated, the most original ones are the minimum average variance estimation (MAVE, Xia et al., 2002) and density-based MAVE (dMAVE, Xia, 2007). These methods do not require the linearity condition and constant variance condition that are critical for inverse regression methods. Ma and Zhu (2012) cast the SDR problem in a semiparametric estimation framework, in which the linearity condition and the constant variance condition on the covariates may be removed but at the cost of performing nonparametric regression. The optimization methods include minimizing general loss functions (Dong & Yang, 2016; Li & Duan, 1989; Soale & Dong, 2020; Wang et al., 2018; Yin & Cook, 2002), and maximizing general dependence measures (Sheng & Yin, 2016; Xue et al., 2017; Yin et al., 2008; Zhang & Chen, 2019). There are

many other methods available in the literature. See (Dong, 2021; Ma & Zhu, 2013) among others for more details.

Until now, we only discuss that the reduced predictors take the linear form $\mathbf{B}^T \mathbf{X}$, where $\mathbf{B}$ is a $p \times q$ matrix that consists of columns $\{\beta_1, \dots, \beta_q\}$. This problem is called linear SDR. However, the predictor and response do not have to be related through linear indices. Nonlinear SDR assumes the reduced predictor taking the nonlinear form $u(\mathbf{X})$, where u is a function of a Hilbert space. The precursors of this topic include PSVM (Li et al., 2011), generalized SIR (GSIR) and generalized SAVE (GSAVE) (Lee et al., 2013), kernel SIR (Wu, 2008; Wu et al., 2013; Yeh et al., 2008), kernel dimension reduction (KDR, Fukumizu et al., 2009) and so on. Li and Song (2017) proposed a general theory for nonlinear SDR where both the predictor and response may be random functions or random vectors. But as far as we know, the existing nonlinear methods are not suitable for survival data.

Survival analysis is an important branch of statistics for analyzing the expected duration of time until one event happens, such as deaths in biological organisms and failures in mechanical systems. Survival data are often censored, and hence it is always challengeable to model such data. Thus, the research on SDR for survival data is not very well developed. DSIR (Double SIR) proposed by Li et al. (1999) is an extension of SIR for survival data. DSIR assumes that the survival time T and the regressor vector $\mathbf{X}$ can be modeled by model (1). Thus, when the linearity condition holds, $E(\mathbf{X}|T) - E(\mathbf{X}) \in \text{span}\{\Sigma\beta_j, j = 1, \dots, q\}$, where $\Sigma = \text{Cov}(\mathbf{X})$ is the covariance matrix of $\mathbf{X}$. For characterizing the information of censoring, DSIR assumes that the censoring time C can be modeled by a model similar to (1). However, DSIR only finds linear SDR directions. Therefore, we need new nonlinear dimension reduction methods for survival data to extract important nonlinear components. The literature on SDR for survival data in recent decades includes Chen (2022), Ding et al. (2020), Kim and Shin (2021), Li and Li (2004), Luand Li (2011), Shevlyakova and Morgenthaler (2014), Sun et al. (2019), Wen (2010), Xia et al. (2010), and Zhao et al. (2022) among others. However, all of them discussed only linear SDR directions. To our knowledge, nonlinear SDR for survival data is a completely new filed.

In this paper, we go one step further to propose a method for nonlinear SDR for survival data. Our method abbreviated as RDSIR (RKHS-based DSIR) is motivated by DSIR and the kernel trick that is an ingenious technique based on the RKHS theory (see Aronszajn, 1950; Berlinet & Thomas-Agnan, 2011; Schölkopf & Smola, 2002; Wahba, 1990 for more details about RKHS). Specifically, we allow the nonlinear function $u(\cdot)$ to reside in an RKHS and an isometric isomorphism is constructed based on the RKHS property. Then $u(\cdot)$ can be represented by the inner product of two elements that reside in the isomorphic feature space. The linear directions found in the isomorphic feature space are corresponding to the nonlinear directions in the original input space.

The rest of the paper is organized as follows. In Section 2, we introduce the nonlinear SDR model for modeling survival data. The implementation of RDSIR is given in Section 3. In Section 4, we investigate the asymptotic properties of RDSIR. In Section 5, we perform Monte Carlo simulations on RDSIR and compare it with DSIR. The performance of RDSIR is illustrated in a real data example. Some concluding remarks are made in Section 6. Due to limited space, all proofs are presented in Data S1.

Throughout this paper, $\mathbb{Z} = \{1, 2, \dots\}$ and $\mathbb{R} = (-\infty, \infty)$. Let $\mathcal{W}$ be a Hilbert space with inner product $\langle \cdot, \cdot \rangle$. For $\alpha, \beta \in \mathcal{W}$, $\alpha \otimes \beta$ is defined by $(\alpha \otimes \beta)\eta = \langle \beta, \eta \rangle \alpha$ for any $\eta \in \mathcal{W}$. Let $\mathcal{W}_1$ and $\mathcal{W}_2$ be two subspaces of $\mathcal{W}$ such that for any $\beta_1 \in \mathcal{W}_1$ and any $\beta_2 \in \mathcal{W}_2$, β_1 is orthogonal to β_2 , then $\mathcal{W}_1$ is said to be orthogonal to $\mathcal{W}_2$. If in addition, any $\beta \in \mathcal{W}$ can be written as a sum of a $\beta_1 \in \mathcal{W}_1$ and a $\beta_2 \in \mathcal{W}_2$, then we write $\mathcal{W} = \mathcal{W}_1 \oplus \mathcal{W}_2$. Let $\mathcal{S}$ be a subspace in a Hilbert space $\mathcal{W}$. $\mathbf{P}_{\mathcal{S}}$ denotes

the orthogonal projection operator on S such that for any $\beta \in \mathcal{W}$, $P_S \beta \in S$ and is orthogonal to $\beta - P_S \beta \in \mathcal{W}$. Let $\mathbf{X}$, $\mathbf{Y}$ and $\mathbf{Z}$ be random vectors. The notation $\mathbf{X} \perp \mathbf{Y}$ means that $\mathbf{X}$ and $\mathbf{Y}$ are independent, while the notation $\mathbf{X} \perp \mathbf{Y} | \mathbf{Z}$ means that $\mathbf{X}$ and $\mathbf{Y}$ are conditionally independent given $\mathbf{Z}$. I_n denotes the $n \times n$ identity matrix, and $\mathbf{1}_n$ stands for the n -dimensional vector $(1, \dots, 1)^T$. Let A be an $n \times m$ matrix with $m \leq n$. Then $P_A = A(A^T A)^{-1} A^T$ is an $n \times n$ projection matrix onto the linear space spanned by columns of A . Let B be a set. $I(B)$ denotes the indicator function of the set B . If Ω is a set, $\#\{\omega \in \Omega\}$ denotes the number of elements in Ω .

2 | NONLINEAR SDR VIA REPRODUCING KERNEL

2.1 | The model setup

Let T , C , and $\mathbf{X} = (X_1, \dots, X_p)^T \in \mathcal{X} \subset \mathbb{R}^p$ which are random, denote, respectively, the true (unobservable) lifetime, the censoring time and the p -dimensional covariates. Put $\tilde{T} = T \wedge C$ and $\Delta = I(T \leq C)$. The commonly made assumption for the censoring mechanism is given below:

Condition 1. $T \perp C | \mathbf{X}$.

Let $\mathcal{H}_R$ be the RKHS induced by the reproducing kernel $\mathcal{R}(\cdot, \cdot)$ with the inner product $\langle \cdot, \cdot \rangle_{\mathcal{H}_R}$ and the norm $\|\cdot\|_{\mathcal{H}_R}$. We assume that the functional dependence of T on $\mathbf{X}$ is given by

$$T = g(u_1(\mathbf{X}), \dots, u_q(\mathbf{X}), \epsilon_T), \quad (2)$$

where g is completely unknown, $u_j(\cdot)$, $1 \leq j \leq q$ are linearly independent in $\mathcal{H}_R$, ϵ_T is the random error with zero mean, and ϵ_T is independent of $\mathbf{X}$.

Remark 1. Let $\sigma(\mathbf{X})$ be the σ -field generated by $\mathbf{X}$. Lee et al. (2013) shown that under mild assumptions, there is a unique sub σ -field $\mathcal{G}^*$ of $\sigma(\mathbf{X})$ such that

1. $Y \perp \mathbf{X} | \mathcal{G}^*$.
2. If $\mathcal{G}$ is a sub σ -field of $\sigma(\mathbf{X})$ such that $Y \perp \mathbf{X} | \mathcal{G}$, then $\mathcal{G}^* \subset \mathcal{G}$.

The σ -field $\mathcal{G}^*$ is called the central σ -field, and the set of functions that are measurable with respect to $\mathcal{G}^*$ is called the central class. In this article, we assume $\{u_1(\mathbf{X}), \dots, u_q(\mathbf{X})\}$ specified in (2) is the central class, that is $\mathcal{G}^* = \sigma\{u_1(\mathbf{X}), \dots, u_q(\mathbf{X})\}$.

We used the word “class” instead of “space” to distinguish these objects from their counterparts in linear SDR. In linear SDR, the central subspace is the space spanned by the coefficient vectors. In the current context, we cannot in general separate u_j from $\mathbf{X}$. Our goal in nonlinear SDR is to estimate the central class. In the subsequent text, we will use the kernel trick to offer a direct connection between nonlinear SDR and the linear SDR. For the sake of discussion, we define the nonlinear central subspace in the feature space.

Let ℓ_2 be the space of all sequences $x = (x_1, x_2, \dots)$ of real numbers satisfying the condition $\sum_{i=1}^{\infty} x_i^2 < \infty$. For $x = (x_1, x_2, \dots)$ and $y = (y_1, y_2, \dots)$ in ℓ_2 , the inner product is defined as $\langle x, y \rangle_2 = \sum_{i=1}^{\infty} x_i y_i$, and the norm of x is $\|x\|_2 = \sqrt{\langle x, x \rangle_2}$. Assume the reproducing kernel $\mathcal{R}(\cdot, \cdot)$ admits the

following spectral decomposition:

$$\mathcal{R}(\mathbf{s}, \mathbf{t}) = \sum_{j=1}^{\infty} a_j \phi_j(\mathbf{s}) \phi_j(\mathbf{t}), \quad \mathbf{s}, \mathbf{t} \in \mathcal{X},$$

where $\{a_j, j \in \mathbb{Z}\}$ is a sequence of nonnegative, non-increasing eigenvalues, $\{\phi_j(\cdot), j \in \mathbb{Z}\}$ is a sequence of corresponding eigenfunctions. The main idea of the kernel trick is first to map the data in the input space $\mathcal{X}$ into the spectrum-based feature space. Define the map $\phi: \mathcal{X} \mapsto \ell_2$ by

$$\phi(\mathbf{s}) = (\sqrt{a_1} \phi_1(\mathbf{s}), \sqrt{a_2} \phi_2(\mathbf{s}), \dots) \in \ell_2, \quad \mathbf{s} \in \mathcal{X}.$$

The ℓ_2 inner product between $\phi(\mathbf{s})$ and $\phi(\mathbf{t})$ is

$$\langle \phi(\mathbf{s}), \phi(\mathbf{t}) \rangle_2 = \mathcal{R}(\mathbf{s}, \mathbf{t}), \quad \mathbf{s}, \mathbf{t} \in \mathcal{X}. \quad (3)$$

It can be shown that the space

$$\mathcal{H} = \overline{\text{span}}\{\phi(\mathbf{s}_1), \phi(\mathbf{s}_2), \dots, \phi(\mathbf{s}_m), \mathbf{s}_j \in \mathcal{X}, j = 1, 2, \dots, m, m \in \mathbb{Z}\},$$

with the inner product (3) is a Hilbert space, where $\overline{A}$ denotes the closure of the set A . Let the map $\mathcal{M}: \mathcal{H}_R \rightarrow \mathcal{H}$ satisfy $\mathcal{M}\mathcal{R}(\mathbf{s}, \cdot) = \phi(\mathbf{s})$. By (3) and the reproducing property of the RKHS, it follows that

$$\langle \mathcal{M}\mathcal{R}(\mathbf{s}, \cdot), \mathcal{M}\mathcal{R}(\mathbf{t}, \cdot) \rangle_2 = \langle \phi(\mathbf{s}), \phi(\mathbf{t}) \rangle_2 = \mathcal{R}(\mathbf{s}, \mathbf{t}) = \langle \mathcal{R}(\mathbf{s}, \cdot), \mathcal{R}(\mathbf{t}, \cdot) \rangle_{\mathcal{H}_R},$$

which implies that $\mathcal{M}$ is an isomorphic map from $\mathcal{H}_R$ to $\mathcal{H}$. Hence, for each $\eta \in \mathcal{H}$, there exists a unique $u(\cdot) \in \mathcal{H}_R$ such that $\mathcal{M}u(\cdot) = \eta$. By the reproducing property of the RKHS, we have

$$u(\mathbf{X}) = \langle u(\cdot), \mathcal{R}(\mathbf{X}, \cdot) \rangle_{\mathcal{H}_R} = \langle \eta, \phi(\mathbf{X}) \rangle_2.$$

Therefore, for $u_1(\cdot), \dots, u_q(\cdot)$ in (2), there exist respectively $\eta_1, \dots, \eta_q$ such that $\mathcal{M}u_j(\cdot) = \eta_j$, $1 \leq j \leq q$. Then model (2) can be written as

$$T = g(\langle \eta_1, \phi(\mathbf{X}) \rangle_2, \dots, \langle \eta_q, \phi(\mathbf{X}) \rangle_2, \epsilon_T). \quad (4)$$

This form is analogous to the linear SDR problem. This model implies that the dependence of T on $\mathbf{X}$ only comes from the q -dimensional variate $S(\mathbf{X}) = (\langle \eta_1, \phi(\mathbf{X}) \rangle_2, \dots, \langle \eta_q, \phi(\mathbf{X}) \rangle_2)$. Although $S(\mathbf{X})$ is a linear statistic in $\mathcal{H}$, it extracts nonlinear features in the original input space $\mathcal{X}$. Since we have assumed that $\{u_1(\mathbf{X}), \dots, u_q(\mathbf{X})\}$ is the central class and there is an isomorphic map from $u_j(\cdot)$ to η_j , it makes sense for us to define $\mathcal{B}_T = \text{span}\{\eta_1, \dots, \eta_q\}$ the nonlinear central subspace. Note that this definition is just for the convenience of developing theory. Our goal is still to estimate the central class $\{u_1(\mathbf{X}), \dots, u_q(\mathbf{X})\}$.

2.2 | The kernel SIR

Let $\mathbf{Z}$ be a random element taking values in $\mathcal{H}$. If $E\|\mathbf{Z}\|^2 < \infty$, the expectation of $\mathbf{Z}$ denoted as $E(\mathbf{Z}) \in \mathcal{H}$, is defined such that $\langle \boldsymbol{\eta}, E(\mathbf{Z}) \rangle = E\langle \boldsymbol{\eta}, \mathbf{Z} \rangle$ for all $\boldsymbol{\eta} \in \mathcal{H}$. The covariance of $\mathbf{Z}$ denoted as $\text{Cov}(\mathbf{Z})$, is given by $E[(\mathbf{Z} - E(\mathbf{Z})) \otimes (\mathbf{Z} - E(\mathbf{Z}))]$. Let P_X denote the probability measure induced by $\mathbf{X}$. Throughout the article, we assume the following condition.

Condition 2. There exists $M > 0$ such that for $\mathbf{X} \in \mathcal{X}$, $\mathcal{R}(\mathbf{X}, \mathbf{X}) \leq M$ with respect to P_X .

Under Condition 2, the random element $\boldsymbol{\phi}(\mathbf{X})$ has a well-defined mean and a covariance operator because $\|\boldsymbol{\phi}(\mathbf{X})\|_2^2 = \mathcal{R}(\mathbf{X}, \mathbf{X})$ is bounded. Besides, Condition 2 also implies that the covariance operator $\Sigma = \text{Cov}(\boldsymbol{\phi}(\mathbf{X}))$ is compact. In fact, Baker and McKeague (1981) pointed out the covariance operator of a strong second-order measure is always compact. Denote $\mu(\boldsymbol{\phi}(\mathbf{X}))$ the probability measure induced by $\boldsymbol{\phi}(\mathbf{X})$. Since $\int_{\mathcal{H}} \|\boldsymbol{\phi}(\mathbf{X})\|_2^2 d\mu(\boldsymbol{\phi}(\mathbf{X})) = \int_{\mathcal{H}} \mathcal{R}(\mathbf{X}, \mathbf{X}) d\mu(\boldsymbol{\phi}(\mathbf{X})) < \infty$, we have $\mu(\boldsymbol{\phi}(\mathbf{X}))$ is a strong-order measure defined in Baker and McKeague (1981). Therefore, the covariance operator $\Sigma = \text{Cov}(\boldsymbol{\phi}(\mathbf{X}))$ is compact with the spectral decomposition

$$\Sigma = \sum_{j=1}^{\infty} v_j \psi_j \otimes \psi_j,$$

where $\{v_1, v_2, \dots\}$ are eigenvalues and $\{\psi_1, \psi_2, \dots\}$ are the corresponding eigenfunctions. The following lemma extends the theoretical foundation of SIR to this nonlinear setting.

Lemma 1. For model (4), if the linearity condition holds, that is for any $\boldsymbol{\eta} \in \mathcal{H}$, there exist $c_0, c_1, \dots, c_q$ such that

$$E(\langle \boldsymbol{\eta}, \boldsymbol{\phi}(\mathbf{X}) \rangle_2 | \langle \boldsymbol{\eta}_1, \boldsymbol{\phi}(\mathbf{X}) \rangle_2, \dots, \langle \boldsymbol{\eta}_q, \boldsymbol{\phi}(\mathbf{X}) \rangle_2) = c_0 + c_1 \langle \boldsymbol{\eta}_1, \boldsymbol{\phi}(\mathbf{X}) \rangle_2 + \dots + c_q \langle \boldsymbol{\eta}_q, \boldsymbol{\phi}(\mathbf{X}) \rangle_2,$$

then $E(\boldsymbol{\phi}(\mathbf{X})|T) - E(\boldsymbol{\phi}(\mathbf{X})) \in \text{span}\{\Sigma \boldsymbol{\eta}_j, j = 1, \dots, q\}$.

Remark 2. As far as we know, the linearity condition is a basic assumption in SIR and nonlinear SDR literature. For example, such an assumption was made in references Wu (2008), Wu et al. (2013), and Yeh et al. (2008). The performance of SIR or KSIR is not very sensitive to this assumption as discussed by Li (1991). Although this assumption is not easy to verify, it is well known that the low-dimensional projection of high-dimensional data often looks like normally distributed. As a result, the linearity condition is approximately true. The proof of Lemma 1 is given in Wu (2008).

By Lemma 1, a basis $\{\beta_1, \dots, \beta_q\}$ of $\mathcal{B}_T$ can be obtained by solving the following generalized eigenvalue-eigenvector problem:

$$\Gamma \beta = \lambda \Sigma \beta, \quad (5)$$

where Γ is the covariance operator of $\boldsymbol{\phi}(\mathbf{X})$ conditional on T , that is, $\Gamma = \text{Cov}(E(\boldsymbol{\phi}(\mathbf{X})|T))$. Under the linearity condition in Lemma 1, Γ is also a compact operator with rank $d_{\Gamma} \leq q$. The spectral

decomposition of Γ is

$$\Gamma = \sum_{j=1}^{d_\Gamma} \zeta_j \varphi_j \otimes \varphi_j, \quad (6)$$

where $\{\zeta_1, \dots, \zeta_{d_\Gamma}\}$ and $\{\varphi_1, \dots, \varphi_{d_\Gamma}\}$ are sequences of eigenvalues and eigenfunctions. The spectral decomposition of Σ and Γ is useful for the development of the asymptotic properties.

In light of (5), we can obtain an estimate of a basis of $\mathcal{B}_T$ by solving the following generalized eigenvalue-eigenvector problem

$$\hat{\Gamma} \beta = \lambda \hat{\Sigma} \beta, \quad (7)$$

where $\hat{\Gamma}$ and $\hat{\Sigma}$ are, respectively, estimates of Γ and Σ . Thus, the remaining problem is to find estimates of Γ and Σ .

Adopting the idea of SIR, Γ is estimated using a simple estimate of the inverse regression curve $E(\phi(\mathbf{X})|T)$ by partitioning the data into several slices according to the Y values and computing the mean of $\phi(\mathbf{X})$ within each slice. Partition $[0, \infty)$ by $0 = t_1 < t_2 < \dots < t_L < t_{L+1} = \infty$. Denote $\mu = E[\phi(\mathbf{X})]$, $\mu_\ell = E[\phi(\mathbf{X})|T \in D_\ell]$ and $p_\ell = P(T \in D_\ell)$, where $D_\ell = [t_\ell, t_{\ell+1})$, $\ell = 1, \dots, L$. Throughout the rest of this paper, we assume that $p_\ell > 0$, for $\ell = 1, \dots, L$. By the definition of μ_ℓ , we have

$$\mu_\ell = \frac{E[\phi(\mathbf{X})I(T \in D_\ell)]}{P\{T \in D_\ell\}} = \frac{E[\phi(\mathbf{X})I(T \geq t_\ell)] - E[\phi(\mathbf{X})I(T \geq t_{\ell+1})]}{E[I(T \geq t_\ell)] - E[I(T \geq t_{\ell+1})]}.$$

Since the observations of T are right censored, we cannot directly estimate $E[\phi(\mathbf{X})I(T \geq t)]$ and $E[I(T \geq t)]$ empirically. However, they can be associated with $\tilde{T}$ and Δ via the following lemma.

Lemma 2. *If Condition 1 holds, then we have*

$$\begin{aligned} E[\phi(\mathbf{X})I(T \geq t)] &= E[\phi(\mathbf{X})I(\tilde{T} \geq t)] + E[\phi(\mathbf{X})I(\tilde{T} < t, \Delta = 0)w(\tilde{T}, t, \phi(\mathbf{X}))], \\ E[I(T \geq t)] &= E[I(\tilde{T} \geq t)] + E[I(\tilde{T} < t, \Delta = 0)w(\tilde{T}, t, \phi(\mathbf{X}))], \end{aligned}$$

where for $t' < t$,

$$w(t', t, \phi(\mathbf{X})) = \frac{S_T(t|\phi(\mathbf{X}))}{S_T(t'|\phi(\mathbf{X}))} = S_T(t|T > t', \phi(\mathbf{X})), \quad (8)$$

$S_T(t|\phi(\mathbf{X})) = P(T \geq t|\phi(\mathbf{X}))$ is the conditional survival function for T given $\phi(\mathbf{X})$. Furthermore, the weight function (8) can be expressed as

$$w(t', t, \phi(\mathbf{X})) = \exp\{-\Lambda(t', t|\phi(\mathbf{X}))\},$$

where

$$\Lambda(t', t|\phi(\mathbf{X})) = E\left\{\frac{I(t' \leq \tilde{T} < t, \Delta = 1)}{S_{\tilde{T}}(\tilde{T}|\phi(\mathbf{X}))} \middle| \phi(\mathbf{X})\right\}, \quad (9)$$

and $S_{\tilde{T}}(\cdot|\phi(\mathbf{X}))$ is the conditional survival function of $\tilde{T}$ conditioned on $\phi(\mathbf{X})$.

Remark 3. We know that T is the unobservable true lifetime, hence we cannot estimate $E[\phi(\mathbf{X})I(T \geq t)]$ and $E[I(T \geq t)]$ directly. Lemma 2 provides a way to estimate them through the conditional survival function $w(t', t, \phi(\mathbf{X}))$. Therefore, the remaining problem is to estimate $w(t', t, \phi(\mathbf{X}))$, or equivalently, to estimate the conditional cumulative hazard function $\Lambda(t', t|\phi(\mathbf{X}))$. By (9), $\Lambda(t', t|\phi(\mathbf{X}))$ is a conditional expectation conditioned on $\phi(\mathbf{X})$ which can be estimated through nonparametric smoothing methods.

2.3 | Double slicing

In this subsection, we present a double slicing method to provide insights for estimating the weight function $w(\cdot, \cdot, \cdot)$. To make full use of the censoring information, we assume that the relationship between C and $\mathbf{X}$ can be modeled by

$$C = h(v_1(\mathbf{X}), \dots, v_c(\mathbf{X}), \epsilon_C), \quad (10)$$

where h is an arbitrary function, $v_j(\cdot)$, $1 \leq j \leq c$, are linearly independent in $\mathcal{H}_R$, and the random error ϵ_C is independent of the covariate $\mathbf{X}$.

Remark 4. Following the same route as Li et al. (1999), we assume the censoring time C satisfies model (10). Some recent approaches, such as Sun et al. (2019) and Zhao et al. (2022) do not require the censoring time C satisfying model (10) and estimation of the censoring distribution to compensate for the bias in estimating the SDR subspace, which offers alternative views on nonlinear SDR for survival data.

Note that with the formulation of (2) and (10), Condition 1 can be reformulated using the conditional independence of the random errors associated with the survival time T and censoring time C , that is $\epsilon_T \perp \epsilon_C | \mathbf{X}$.

Similar to (4), model (10) can be expressed as

$$C = h(\langle \gamma_1, \phi(\mathbf{X}) \rangle_2, \dots, \langle \gamma_c, \phi(\mathbf{X}) \rangle_2, \epsilon_C), \quad (11)$$

where $\gamma_j \in \mathcal{H}$, $j = 1, \dots, c$. Let $\mathcal{B}_C$ denote the central subspace spanned by $\{\gamma_1, \dots, \gamma_c\}$. Let $\mathcal{B}_T \subset \mathcal{H}$ be the smallest Hilbert space containing both $\mathcal{B}_T$ and $\mathcal{B}_C$. Then the dimension of $\mathcal{B}_T$ is at most $q + c$. By (4), we have $T \perp \mathbf{X} | \mathcal{P}_{\mathcal{B}_T} \phi(\mathbf{X})$. Thus, from the definition of $w(t', t, \phi(\mathbf{X}))$ given in (8), $w(t', t, \phi(\mathbf{X}))$ can be rewritten as

$$w(t', t, \phi(\mathbf{X})) = \frac{P(T > t | \phi(\mathbf{X}))}{P(T > t' | \phi(\mathbf{X}))} = \frac{P(T > t | \mathcal{P}_{\mathcal{B}_T} \phi(\mathbf{X}))}{P(T > t' | \mathcal{P}_{\mathcal{B}_T} \phi(\mathbf{X}))} = w(t', t, \mathcal{P}_{\mathcal{B}_T} \phi(\mathbf{X})). \quad (12)$$

Since $E(\phi(\mathbf{X}) | \tilde{T}, \Delta) = E(E(\phi(\mathbf{X}) | T, C) | \tilde{T}, \Delta)$, under the condition of Lemma 1, it can be shown in the same way as in Lemma 1 that

$$E(\phi(\mathbf{X}) | \tilde{T}, \Delta) - E(\phi(\mathbf{X})) \in \text{span}\{\Sigma \eta_1, \dots, \Sigma \eta_q, \Sigma \gamma_1, \dots, \Sigma \gamma_c\}.$$

Thus, a basis of $\mathcal{B}_J$ can be found by solving the following generalized eigenvalue–eigenvector problem

$$\Gamma_J \beta = \lambda_J \Sigma \beta, \quad (13)$$

where $\Gamma_J = \text{Cov}(\psi_J(\tilde{T}, \Delta))$ with $\psi_J(\tilde{T}, 0) = E(\phi(\mathbf{X})|\tilde{T}, \Delta = 0)$ and $\psi_J(\tilde{T}, 1) = E(\phi(\mathbf{X})|\tilde{T}, \Delta = 1)$.

Remark 5. Here, we use $E(\phi(\mathbf{X})|\tilde{T}, \Delta) - E(\phi(\mathbf{X}))$ to recover the central subspace for (T, C) denoted as S . It extracts information about the central subspace via inverse regression lines and recovers at least a portion of the central subspace. To be more specific, we define the inverse regression subspace as

$$S_{E(\phi(\mathbf{X})|Y)} = \text{span}\{E(\phi(\mathbf{X})|Y) - E(\phi(\mathbf{X}))|Y \in \mathbb{R}\}.$$

The inverse regression subspace is spanned by the centered inverse regression curve $E(\phi(\mathbf{X})|Y) - E(\phi(\mathbf{X}))$ as Y varies. Under the theoretical framework and notation system of this article, we have $S_{E(\phi(\mathbf{X})|T, C)} \subset S$. However, (T, C) is not observable and only $(\tilde{T}, \Delta)$ is observed. Since $E(\phi(\mathbf{X})|\tilde{T}, \Delta) = E(E(\phi(\mathbf{X})|T, C)|\tilde{T}, \Delta)$, it follows that

$$S_{E(\phi(\mathbf{X})|\tilde{T}, \Delta)} \subset S_{E(\phi(\mathbf{X})|T, C)} \subset S.$$

Thus, we can use $S_{E(\phi(\mathbf{X})|\tilde{T}, \Delta)}$ as an approximation of S . It is obvious that $S_{E(\phi(\mathbf{X})|\tilde{T})}$ and $S_{E(\phi(\mathbf{X})|\Delta)}$ are both in S . However, they both are poor estimates of S . The σ -algebra generated by $\tilde{T}$ or Δ is contained in the σ -algebra generated by $(\tilde{T}, \Delta)$. In fact, we have $S_{E(\phi(\mathbf{X})|\tilde{T})} \subset S_{E(\phi(\mathbf{X})|\tilde{T}, \Delta)}$ and $S_{E(\phi(\mathbf{X})|\Delta)} \subset S_{E(\phi(\mathbf{X})|\tilde{T}, \Delta)}$.

2.4 | Estimation

Let $\{\tilde{T}_i, \Delta_i, \mathbf{X}_i, i = 1, \dots, n\}$ be the n i.i.d. observations from the distribution of $(\tilde{T}, \Delta, \mathbf{X})$. Without loss of generality, we assume $\mu = E(\phi(\mathbf{X})) = 0$. Denote the centered mapped data by $\Phi = (\phi(\mathbf{X}_1), \dots, \phi(\mathbf{X}_n))$. By Lemma 2, the estimates of $E[\phi(\mathbf{X})I(T \geq t)]$ and $E[I(T \geq t)]$ can be constructed as follows:

$$\begin{aligned} \hat{E}[\phi(\mathbf{X})I(T \geq t)] &= \frac{1}{n} \sum_{i: \tilde{T}_i \geq t} \phi(\mathbf{X}_i) + \frac{1}{n} \sum_{i: \tilde{T}_i < t, \Delta_i = 0} \phi(\mathbf{X}_i) \hat{w}(\tilde{T}_i, t, \phi(\mathbf{X}_i)), \\ \hat{P}\{T \geq t\} &= \frac{1}{n} \sum_{i=1}^n I(\tilde{T}_i \geq t) + \frac{1}{n} \sum_{i: \tilde{T}_i < t, \Delta_i = 0} \hat{w}(\tilde{T}_i, t, \phi(\mathbf{X}_i)), \end{aligned}$$

where $\hat{w}(\cdot, \cdot, \cdot)$ is an estimate of the weight function $w(\cdot, \cdot, \cdot)$ to be discussed in detail later in (24). Denote

$$\hat{w}_{i\ell} = \begin{cases} \hat{w}(\tilde{T}_i, t_\ell, \phi(\mathbf{X}_i)) - \hat{w}(\tilde{T}_i, t_{\ell+1}, \phi(\mathbf{X}_i)), & \Delta_i = 0, \\ I(t_\ell \leq \tilde{T}_i < t_{\ell+1}), & \Delta_i = 1, \end{cases}$$

then

$$\begin{aligned}\hat{E}[\phi(\mathbf{X})I(t_\ell \leq T < t_{\ell+1})] &= \frac{1}{n}\Phi\hat{\mathbf{w}}_\ell, \\ \hat{p}_\ell &= \hat{P}\{T \geq t_\ell\} - \hat{P}\{T \geq t_{\ell+1}\} = \frac{1}{n}\mathbf{1}_n^T\hat{\mathbf{w}}_\ell,\end{aligned}$$

where $\hat{\mathbf{w}}_\ell = (\hat{w}_{1\ell}, \dots, \hat{w}_{n\ell})^T$. Thus, an estimator of μ_ℓ is given by

$$\hat{\mu}_\ell = \frac{\hat{E}[\phi(\mathbf{X})I(t_\ell \leq T < t_{\ell+1})]}{\hat{p}_\ell} = \frac{\frac{1}{n}\Phi\hat{\mathbf{w}}_\ell}{\hat{p}_\ell} = \frac{\Phi\hat{\mathbf{w}}_\ell}{\mathbf{1}_n^T\hat{\mathbf{w}}_\ell}.$$

Then we can estimate Γ and Σ by

$$\hat{\Sigma} = \frac{1}{n} \sum_{i=1}^n \phi(\mathbf{X}_i) \otimes \phi(\mathbf{X}_i) = \frac{1}{n} \Phi \Phi^T, \quad (14)$$

$$\hat{\Gamma} = \sum_{\ell=1}^L \hat{p}_\ell \hat{\mu}_\ell \otimes \hat{\mu}_\ell = \frac{1}{n} \Phi Q \Phi^T, \quad (15)$$

where $Q = (1/n)\hat{W}\hat{P}^{-1}\hat{W}^T$, $\hat{W} = (\hat{\mathbf{w}}_1, \dots, \hat{\mathbf{w}}_L)$, $\hat{P} = \text{diag}(\hat{p}_1, \dots, \hat{p}_L)$. Since $\mathcal{B}_T$ is the central subspace, by (14) and (15), the solution of (7) should be in the space spanned by $\{\phi(\mathbf{X}_1), \dots, \phi(\mathbf{X}_n)\}$. The following lemma gives a representation of the solution of (7).

Lemma 3. *Solutions of (7) which is the empirical version of the infinite-dimensional problem (5) are as follows:*

$$\hat{\beta}_j = \sum_{i=1}^n \hat{\alpha}_j^{(i)} \phi(\mathbf{X}_i) = \Phi \hat{\alpha}_j, \quad j = 1, \dots, q, \quad (16)$$

where $\hat{\alpha}_j = (\hat{\alpha}_j^{(1)}, \dots, \hat{\alpha}_j^{(n)})^T$, $j = 1, \dots, q$. In addition, $\hat{\alpha}_1, \dots, \hat{\alpha}_q$ are the solutions of the following finite-dimensional generalized eigenvalue–eigenvector problem:

$$RQR\alpha = \lambda R^2\alpha, \quad (17)$$

where $R = \Phi^T\Phi$. Similar results hold true for using the following regularization technique:

$$\hat{\Sigma}\hat{\Gamma}\beta = \lambda(\hat{\Sigma}^2 + \tau I_n)\beta, \quad (18)$$

that is, the solutions of (18) have the same type of expressions as (16) with $\hat{\alpha}_j$, $j = 1, \dots, q$ being the solutions of

$$RQR\alpha = \lambda(R^2 + n^2\tau I_n)\alpha, \quad (19)$$

where $\tau > 0$ is a regularization parameter.

Remark 6. The proof of Lemma 3 is similar to that of property 7 in Wu et al. (2013). Instead of considering complete data as in Wu et al. (2013), we deal with censored survival data and consider the information of censoring contained in Q in this paper. The representation (16) is of great importance since it converts an infinite-dimensional problem (7) to a finite-dimensional one (17) and hence the proposed methodology is applicable. Similar to the methods in Li and Yin (2008) and Zhong et al. (2005), we add a regularization term to the right-hand side of (17) to alleviate over-fitting and numerically instability, which results in (19). The selection of the tuning parameter τ will be elaborated in Section 3.

The relationship between the kernel matrix of centered mapped data and that of raw mapped data is known. As discussed in Wu (2008), this relationship is

$$R = \left(I_n - \frac{1}{n} \mathbf{1}_n \mathbf{1}_n^T \right) \tilde{R} \left(I_n - \frac{1}{n} \mathbf{1}_n \mathbf{1}_n^T \right),$$

where $\tilde{R} = \mathcal{R}(\mathbf{X}_i, \mathbf{X}_j)_{1 \leq i, j \leq n}$ is the Gram matrix of $\mathcal{R}(\cdot, \cdot)$ over $\mathbf{X}_1, \dots, \mathbf{X}_n$.

Next we use the methods in Li et al. (1999) to obtain the empirical version of (13). We partition the interval $[0, \infty)$ for the censored and uncensored lifetime as follows:

$$0 = t_1^{[l]} < t_2^{[l]} < \dots < t_{L_l}^{[l]} < t_{L_l+1}^{[l]} = \infty, \quad l = 0, 1,$$

where L_l denotes the number of partitions, and $l = 0, 1$ refer to the two different partitions to be applied to the censored and uncensored lifetime. Put $D_\ell^{[l]} = [t_\ell^{[l]}, t_{\ell+1}^{[l]})$, and $n_{l,\ell} = \#\{i : \Delta_i = l, \tilde{T}_i \in D_\ell^{[l]}\}$. The corresponding ordered time in $D_\ell^{[l]}$ consists of $\tilde{T}_{1,\ell}^{[l]}, \dots, \tilde{T}_{n_{l,\ell},\ell}^{[l]}$. Denote $\Phi_\ell^{[l]} = (\phi(\mathbf{X}_{1,\ell}^{[l]}), \dots, \phi(\mathbf{X}_{n_{l,\ell},\ell}^{[l]}))$ and $\mu_\ell^{[l]} = E(\phi(\mathbf{X}) | \Delta = l, \tilde{T} \in D_\ell^{[l]})$. The empirical version of $\mu_\ell^{[l]}$ is given by

$$\hat{\mu}_\ell^{[l]} = \frac{1}{n \hat{p}_\ell^{[l]}} \sum_{i=1}^n \phi(\mathbf{X}_i) I(\Delta_i = l, \tilde{T}_i \in D_\ell^{[l]}), \quad l = 0, 1,$$

where $\hat{p}_\ell^{[l]} = n_{l,\ell}/n$. Thus, Γ_J can be estimated by

$$\hat{\Gamma}_J = \sum_{l \in \{0,1\}} \sum_{\ell=1}^{L_l} \hat{p}_\ell^{[l]} \hat{\mu}_\ell^{[l]} \otimes \hat{\mu}_\ell^{[l]} = \frac{1}{n} \Phi Q_J \Phi^T, \quad (20)$$

where $Q_J = BAB^T$, $A = \text{diag}(P_{1_{n_{0,1}}}, \dots, P_{1_{n_{0,L_0}}}, P_{1_{n_{1,1}}}, \dots, P_{1_{n_{1,L_1}}})$, B is an $n \times n$ permutation matrix such that $\tilde{\Phi} = \Phi B$ and $\tilde{\Phi} = (\Phi_1^{[0]}, \dots, \Phi_{L_0}^{[0]}, \Phi_1^{[1]}, \dots, \Phi_{L_1}^{[1]})$. Thus, an empirical version of (13) is

$$\hat{\Gamma}_J \beta = \lambda_J \hat{\Sigma} \beta. \quad (21)$$

Denote $\mathcal{H}_1 = \text{span}\{\phi(\mathbf{X}_1), \dots, \phi(\mathbf{X}_n)\}$ and $\mathcal{H}_2$ to be the orthogonal complement subspace of $\mathcal{H}_1$, that is, $\mathcal{H}_2 \perp \mathcal{H}_1$ such that $\mathcal{H} = \mathcal{H}_1 \oplus \mathcal{H}_2$. In view of (14) and (20), for any $\beta \in \mathcal{H}_2$, $\hat{\Gamma}_J \beta = \lambda_J \hat{\Sigma} \beta = 0$. Since we seek a basis for $\mathcal{B}_J$ which is the smallest Hilbert space containing both $\mathcal{B}_T$ and $\mathcal{B}_C$, so the solutions of (21) should be in $\mathcal{H}_1$. Denote the set of all such solutions corresponding to

the nonzero λ_J by $\{\hat{\beta}_{J1}, \dots, \hat{\beta}_{Jm}\}$. Since $\hat{\beta}_{Jl} \in \mathcal{H}_1$, there exists $\hat{\alpha}_{Jl}$ such that $\hat{\beta}_{Jl} = \Phi \hat{\alpha}_{Jl}$ for $1 \leq l \leq m$. Results of Lemma 3 can be extended to the case corresponding to joint SDR. When Q is replaced by Q_J , similar results can be obtained. By (14), (20), and the fact that $R = \Phi^T \Phi$, (21) is equivalent to

$$RQ_J R \alpha = \lambda_J R^2 \alpha. \quad (22)$$

As done in Lemma 2, we modify (22) by using the following regularization technique

$$RQ_J R \alpha = \lambda_J (R^2 + n^2 s I_n) \alpha, \quad (23)$$

where s is a tuning parameter, which can be selected by cross-validation. Denote the first m significant solutions of (23) by $\hat{A}_J = (\hat{\alpha}_{J1}, \dots, \hat{\alpha}_{Jm})$. Then we obtain $\{\hat{\beta}_{J1}, \dots, \hat{\beta}_{Jm}\}$, where $\hat{\beta}_{Jl} = \Phi \hat{\alpha}_{Jl}$ for $1 \leq l \leq m$. Define $R(X_{\leq n}, X) = (\mathcal{R}(X_1, X), \dots, \mathcal{R}(X_n, X))^T$. We have

$$\langle \hat{\beta}_{Jl}, \phi(X) \rangle_2 = \langle \Phi \hat{\alpha}_{Jl}, \phi(X) \rangle_2 = (\Phi \hat{\alpha}_{Jl})^T \phi(X) = R(X_{\leq n}, X)^T \hat{\alpha}_{Jl}, \quad l = 1, \dots, m.$$

In terms of (12), $\phi(X)$ can be replaced by $P_{B_J} \phi(X)$ in (9), that is, $\Lambda(t', t | \phi(X)) = \Lambda(t', t | P_{B_J} \phi(X))$. For simplicity, we use kernel smoothing method to estimate the conditional cumulative hazard function $\Lambda(t', t | \phi(X))$. Let $K_m(\cdot)$ be a multivariate kernel function defined on $\mathbb{R}^m$ and h_n be the bandwidth in each coordinate. We consider K_m taking a product form, $K_m(\mathbf{x}) = K(\mathbf{x}^{(1)}) \cdots K(\mathbf{x}^{(m)})$ for some one-dimensional kernel function $K(\cdot)$, with $\mathbf{x}^{(i)}$ being the i -th component of the m -dimensional vector $\mathbf{x}$. To alleviate the inefficiency caused by the curse of dimensionality, we use the density of $P_{B_J} \phi(X)$ as a threshold to determine where over a data range multivariate smoothing is reliable (Taylor & Einbeck, 2013). The basic idea is to identify regions which are suitable for local regression estimation by looking at the density of the covariates. Denote the estimate of the density of $P_{B_J} \phi(X)$ as $\hat{f}(P_{B_J} \phi(X))$. A threshold d is sought such that, if at a point X_i , $\hat{f}(P_{B_J} \phi(X_i)) > d$, then an estimate using local regression can be considered somewhat reliable. Follow the recommendation of Taylor and Einbeck (2013), we take $d = \rho_m K_m(\mathbf{0}) n^{-1} h_n^{-m/2}$, where ρ_m is a constant that depends on m , for example $\rho_3 = 2.19$, $\rho_4 = 3.12$, and $\rho_5 = 4.46$. The bandwidth h_n can be obtained through standard routines such as Scott's rule (Scott, 2015). With a slight abuse of notation, we still denote the corresponding sample points that satisfy $\hat{f}(P_{B_J} \phi(X_i)) > d$ as $\{\tilde{T}_i, \Delta_i, X_i, i = 1, \dots, n\}$. The kernel estimate of $\Lambda(t', t | P_{B_J} \phi(X))$ is

$$\hat{\Lambda}(t', t | P_{B_J} \phi(X)) = \frac{n^{-1} \sum_{i: \substack{t' < \tilde{T}_i < t, \\ \Delta_i = 1}} [\hat{S}_T(\tilde{T}_i | P_{B_J} \phi(X_i))^{-1} h_n^{-m} K_m(h_n^{-1} [R(X_{\leq n}, X_i) - R(X_{\leq n}, X)]^T \hat{A}_J)]}{\hat{f}(P_{B_J} \phi(X))},$$

where

$$\hat{S}_T(\tilde{T}_i | P_{B_J} \phi(X_i)) = \frac{n^{-1} \sum_{j: \tilde{T}_j > \tilde{T}_i} h_n^{-m} K_m(h_n^{-1} (R(X_{\leq n}, X_j) - R(X_{\leq n}, X_i))^T \hat{A}_J)}{\hat{f}(P_{B_J} \phi(X_i))},$$

$$\hat{f}(P_{B_J} \phi(X)) = n^{-1} \sum_{j=1}^n h_n^{-m} K_m(h_n^{-1} (R(X_{\leq n}, X_j) - R(X_{\leq n}, X))^T \hat{A}_J).$$

By the estimation of Λ , we have

$$\hat{w}(t', t, \mathbf{P}_{B_j} \boldsymbol{\phi}(\mathbf{X})) = \exp\{\hat{\Lambda}(t', t | \mathbf{P}_{B_j} \boldsymbol{\phi}(\mathbf{X}))\}. \quad (24)$$

Then we can solve the generalized eigenvalue–eigenvector problem (19) to get $\{\hat{\boldsymbol{\alpha}}_j, j = 1, 2, \dots, q\}$. The estimate of the central class is

$$\hat{u}_j(\mathbf{X}) = \langle \hat{\boldsymbol{\beta}}_j, \boldsymbol{\phi}(\mathbf{X}) \rangle_2 = R(\mathbf{X}_{\leq n}, \mathbf{X})^T \hat{\boldsymbol{\alpha}}_j, \quad j = 1, \dots, q,$$

where $\hat{\boldsymbol{\beta}}_j = \Phi \hat{\boldsymbol{\alpha}}_j$.

3 | IMPLEMENTATION OF THE PROPOSED METHOD

In the following, we present an algorithm for implementing the proposed method.

Step 1 Chose a reproducing kernel $\mathcal{R}(\cdot, \cdot)$ and calculate the Gram matrix R .

Step 2 Apply double slicing on $(\tilde{T}, \Delta)$ to compute Q_J . Then solve

$$RQ_J R \boldsymbol{\alpha} = \lambda_J (R^2 + n^2 s I_n) \boldsymbol{\alpha},$$

to obtain the first m significant eigenvectors $\hat{\boldsymbol{\alpha}}_{J1}^0, \dots, \hat{\boldsymbol{\alpha}}_{Jm}^0$, and unitize them to $\hat{A}_J = (\hat{\boldsymbol{\alpha}}_{J1}, \dots, \hat{\boldsymbol{\alpha}}_{Jm})$ where

$$\hat{\boldsymbol{\alpha}}_{Jj} = \hat{\boldsymbol{\alpha}}_{Jj}^0 / \sqrt{\hat{\boldsymbol{\alpha}}_{Jj}^{0T} R \hat{\boldsymbol{\alpha}}_{Jj}^0},$$

satisfying that $\langle \hat{\boldsymbol{\beta}}_{Jj}, \hat{\boldsymbol{\beta}}_{Jj} \rangle_2 = \langle \Phi \hat{\boldsymbol{\alpha}}_{Jj}, \Phi \hat{\boldsymbol{\alpha}}_{Jj} \rangle_2 = \hat{\boldsymbol{\alpha}}_{Jj}^T R \hat{\boldsymbol{\alpha}}_{Jj} = 1$.

Step 3 First apply the m -dimensional kernel smoothing to obtain the weight function $\hat{w}$, and then calculate $\hat{W}$, $\hat{P}$ and Q . Afterwards, solve

$$RQR \boldsymbol{\alpha} = \lambda (R^2 + n^2 \tau I_n) \boldsymbol{\alpha},$$

to obtain the first q significant eigenvectors $\hat{\boldsymbol{\alpha}}_1, \dots, \hat{\boldsymbol{\alpha}}_q$ and then compute

$$\hat{u}_j(\mathbf{X}) = R(\mathbf{X}_{\leq n}, \mathbf{X})^T \hat{\boldsymbol{\alpha}}_j, \quad j = 1, \dots, q.$$

In our simulation study in Section 5.1, we set the $L = L_0 = L_1 = 10$ that is a moderate value for the number of slicing (Li, 1991). It is noted that m and q , the numbers of significant eigenvectors, are decided in the same way as in principal component analysis. Here, we take the eigenvectors corresponding to the first m (or q) eigenvalues that capture 95% of the total sum of the eigenvalues.

It is important to select a proper regularization parameter τ or s as it controls the tradeoff between the bias and the variance of the estimator. Similar to Zhong et al. (2005), we consider the mean squared error of the SDR directions

$$L(\tau) = V(\tau) + B(\tau) = \sum_{i=1}^n \sum_{j=1}^q \text{var}(\hat{u}_j(\mathbf{X}_i)(\tau)) + \sum_{i=1}^n \sum_{j=1}^q (E(\hat{u}_j(\mathbf{X}_i)(\tau)) - u_j(\mathbf{X}_i))^2, \quad (25)$$

where $V(\tau)$ is the sum of the variances of $\hat{u}_j(\mathbf{X}_i)(\tau)$, $i = 1, \dots, n$, $j = 1, \dots, q$, $B(\tau)$ is the corresponding sum of the squared biases. Here, we use the notation $\hat{u}_j(\mathbf{X}_i)(\tau)$ instead of $\hat{u}_j(\mathbf{X}_i)$ to emphasize the dependence of $\hat{u}_j(\mathbf{X}_i)$ on τ . We choose τ that minimizes $L(\tau)$. However, $L(\tau)$ is not directly computed since both $V(\tau)$ and $B(\tau)$ cannot be computed directly. To solve this problem, we can turn to a bootstrap procedure. This strategy is motivated by Zhong et al. (2005). Specifically, we denote $\hat{u}_1^{(b)}(\mathbf{X})(\tau), \dots, \hat{u}_q^{(b)}(\mathbf{X})(\tau)$ obtained from the b th bootstrap sample, $b = 1, \dots, B$. The estimates of $E(\hat{u}_j(\mathbf{X}_i)(\tau))$ and $\text{var}(\hat{u}_j(\mathbf{X}_i)(\tau))$ are thus given by

$$\hat{E}(\hat{u}_j(\mathbf{X}_i)(\tau)) = \frac{1}{B} \sum_{b=1}^B \hat{u}_j^{(b)}(\mathbf{X}_i)(\tau),$$

$$\hat{\text{var}}(\hat{u}_j(\mathbf{X}_i)(\tau)) = \frac{1}{B-1} \sum_{b=1}^B \left(\hat{u}_j^{(b)}(\mathbf{X}_i)(\tau) - \hat{E}(\hat{u}_j(\mathbf{X}_i)(\tau)) \right)^2.$$

In practice, the true direction $u_j(\mathbf{X}_i)$ is also unknown but it can be approximated by the bootstrap mean of $E(\hat{u}_j(\mathbf{X}_i)(\tau_0))$, where τ_0 is a suitably chosen small positive value. Similar to Li and Song (2017), we recommend to choose τ_0 such that $n^2\tau_0 = 0.05\lambda_1(R^2)$, where $\lambda_1(R^2)$ denotes the largest eigenvalue of R^2 . We minimize $L(\tau)$ over a grid of τ in $[\tau_0/20, 20\tau_0]$. The grid consists of 20 points, equally spaced in log scale. Although the proposed procedure for determining τ is based on an approximation, our experience from the extensive simulations suggests that it works well.

4 | ASYMPTOTIC PROPERTIES

In this section, we study the asymptotic properties of the SDR directions estimated by RDSIR. We first need to show that the following two conditions hold true:

$$(L1) \quad |\hat{p}_\ell - p_\ell| = O_p(n^{-\kappa}),$$

$$(L2) \quad \left\| \frac{1}{n} \Phi \hat{\mathbf{w}}_\ell - E[\phi(\mathbf{X})I(T \in D_\ell)] \right\|_2 = O_p(n^{-\kappa}),$$

where $\kappa \in (0, 1/4)$. We note that $\hat{p}_l = \hat{P}(T \geq t_l) - \hat{P}(T \geq t_{l+1}) = \frac{1}{n} \mathbf{1}_n^T \hat{\mathbf{w}}_\ell$. To prove (L1), we need only to show

$$|\hat{P}(T \geq t) - P(T \geq t)| = O_p(n^{-\kappa}).$$

We denote a basis of B_J by $B_J = \{\beta_{J1}, \dots, \beta_{Jm}\}$, which is found via the generalized eigenvalue–eigenvector problem (13), and its corresponding estimator by $\hat{B}_J = \{\hat{\beta}_{J1}, \dots, \hat{\beta}_{Jm}\}$ that is obtained via the regularization problem (23). To simplify notations, we denote $\mathbf{Z}_i = B_J^T \phi(\mathbf{X}_i)$ and $\hat{\mathbf{Z}}_i = \hat{B}_J^T \phi(\mathbf{X}_i)$, where $\mathbf{Z}_i$ is an m -dimensional vector whose k th component is $\mathbf{Z}_i^{(k)} = \langle \beta_{Jk}, \phi(\mathbf{X}_i) \rangle$, and $\hat{\mathbf{Z}}_i$ is the corresponding estimator of $\mathbf{Z}_i$. To further simplify the notations, let $e_{ij} = h_n^{-m} K_m(h_n^{-1}(\mathbf{Z}_i - \mathbf{Z}_j))$, $u_{ij} = e_{ij} - E(e_{ij}|\mathbf{Z}_j)$, $v_{kj} = I(\tilde{T}_k > \tilde{T}_j)e_{kj} - E[I(\tilde{T}_k > \tilde{T}_j)e_{kj}|\mathbf{Z}_j, \tilde{T}_j]$, $f_j = f(\mathbf{Z}_j)$, $S_j = S_{\tilde{T}}(\tilde{T}_j|\mathbf{Z}_j)$, $I_{ij} = I(\tilde{T}_i < \tilde{T}_j < t, \Delta_j = 1)$, $\Lambda_i = \Lambda(\tilde{T}_i, t|\mathbf{Z}_i)$, $\hat{\Lambda}_i = \hat{\Lambda}(\tilde{T}_i, t|\mathbf{Z}_i)$ and $w_i = e^{-\Lambda_i}$. We make the following assumptions:

$$(A1) \quad E(e_{ij}|\mathbf{Z}_j) - f(\mathbf{Z}_j) = O_p(n^{-1/2}).$$

- (A2) $\frac{1}{n} \sum_k u_{ki} = O_p(n^{-1/4})$.
- (A3) $E[I(\tilde{T}_k > \tilde{T}_j)e_{kj}|\mathbf{Z}_j, \tilde{T}_j] - S_{\tilde{T}}(\tilde{T}_j|\mathbf{Z}_j)f(\mathbf{Z}_j) = O_p(n^{-1/2})$.
- (A4) $\frac{1}{n} \sum_k v_{ki} = O_p(n^{-1/4})$.
- (A5) $E(I_{ij}S_j^{-1}e_{ji}|\mathbf{Z}_i, \tilde{T}_i) - \Lambda_{if_i} = O_p(n^{-1/2})$.
- (A6) $\int x^i K(x)dx = 0$, $i = 1, \dots, d-1$ and $\int x^d K(x)dx \neq 0$ where $d > (m+1)/b$ with $0 < b \leq 1/4$. $f(\cdot)$ and $S(\cdot|\mathbf{Z}_j)$ are d -times continuously differentiable and bounded away from zero.
- (A7) The tuning parameter $s = s(n)$ in (23) satisfies $\lim_{n \rightarrow \infty} s = 0$ and $\lim_{n \rightarrow \infty} s\sqrt{n} = \infty$.
- (A8) $\{\beta_{jj}\}_{j=1}^m$ depends only on finite eigenfunctions of Σ .

Assumptions (A1)–(A5) are the regularity conditions that are the parallel extensions of the assumptions made in lemma 3.1 of Li et al. (1999) and are satisfied with the bandwidth $h_n \propto n^{-1/2d}$ provided $2m \leq d$. Assumption (A1) implies that the bias term of $\hat{f}(\mathbf{Z}_i)$ is of root n rate which requires a bandwidth smaller than the usual optimal one. By standard Taylor expansion,

$$E(e_{ij}|\mathbf{Z}_j) - f(\mathbf{Z}_j) = C(\mathbf{Z}_j, K_m, d)h_n^d + o(h_n^d),$$

where $C(\mathbf{Z}_j, K_m, d)$ is a constant related to $\mathbf{Z}_j$, K_m and d . Thus, when $h_n \propto n^{-1/2d}$, we have $E(e_{ij}|\mathbf{Z}_j) - f(\mathbf{Z}_j) = O_p(n^{-1/2})$. With (A1), we can write

$$\hat{f}(\mathbf{Z}_i) = f(\mathbf{Z}_i) + \frac{1}{n} \sum_k u_{ki} + O_p(n^{-1/2}).$$

Here, if we treat $\mathbf{Z}_j$ as fixed, then by central limit theorem, $n^{-1} \sum_k u_{ki} = O_p(n^{-1/2+m/2d})$. Assumption (A2) is satisfied when $d \geq 2m$. Assumption (A2) is a mild and flexible one which only requires that the rate of the term $\sum_k u_{ki}/n$ contributing to the variance of $\hat{f}(\mathbf{Z}_i)$ is $O_p(n^{-1/4})$. Assumptions (A3) and (A5) are made so that the biases of both kernel estimates of $S_{\tilde{T}f_j}$ and Λ_{if_i} have the root n rate. With suitable smoothness conditions on conditional survival function $S_{\tilde{T}}(t|\mathbf{Z})$ and cumulative hazard function $\Lambda(t', t|\mathbf{Z})$, the bandwidth to achieve Assumption (A1) may also imply that both Assumptions (A3) and (A5) hold true. Assumption (A4) is made for the same reason as Assumption (A2). Assumption (A6) is imposed on the kernel function, which is similar to those made in Li et al. (1999), which also ensures the rationality of the results. It is noted that a relatively large d can be chosen such that $\kappa = b - (m+1)/d > 0$. We can refer to Jones and Foster (1993) for the construction of this kind of higher order kernel. Assumptions (A7)–(A8) that are the same assumptions in theorem 9 of Wu et al. (2013), which have been used to study the convergence rates of the joint SDR directions for both survival and censoring times there. Under Assumptions (A7) and (A8), we can learn from the results of Wu et al. (2013) that $\hat{\mathbf{Z}}_i^{(k)} - \mathbf{Z}_i^{(k)} = O_p(n^{-b})$ where $0 < b \leq 1/4$. We can now formally introduce the following important lemma.

Lemma 4. Under Assumptions (A1)–(A8), there exists a positive number $\kappa \in (0, 1/4)$ such that

$$(L1) \quad |\hat{p}_\ell - p_\ell| = O_p(n^{-\kappa}),$$

$$(L2) \quad \left\| \frac{1}{n} \Phi \hat{\mathbf{w}}_\ell - E[\phi(\mathbf{X})I(T \in D_\ell)] \right\| = O_p(n^{-\kappa}).$$

Our asymptotic results are based on the perturbation theory for linear operators. We first introduce some properties of Hilbert–Schmidt operators. Let $\mathcal{H}$ be a Hilbert space with norm $\|\cdot\|$ and $\{\xi_i, i \in I\}$ a standard orthogonal basis of $\mathcal{H}$. We say that linear operator $\mathcal{L}$ defined on $\mathcal{H}$ is a Hilbert–Schmidt operator if

$$\|\mathcal{L}\|_{\text{HS}}^2 = \sum_{i=1}^{\infty} \|\mathcal{L}\xi_i\|_{\mathcal{H}}^2 < \infty.$$

Hilbert–Schmidt class spans as a new Hilbert space with $\|\cdot\|_{\text{HS}}$ norm. If S is a bounded operator in $\mathcal{H}$, then $S\mathcal{L}$ and $\mathcal{L}S$ belongs to Hilbert–Schmidt class and

$$\|S\mathcal{L}\|_{\text{HS}} \leq \|S\|_{\text{op}} \|\mathcal{L}\|_{\text{HS}}, \quad \|\mathcal{L}S\|_{\text{HS}} \leq \|\mathcal{L}\|_{\text{op}} \|S\|_{\text{HS}},$$

where $\|\cdot\|_{\text{op}}$ is the operator norm

$$\|\mathcal{L}\|_{\text{op}} = \sup_{f \in \mathcal{H}} \frac{\|\mathcal{L}f\|}{\|f\|}.$$

Recall that $\Sigma = \sum_{j=1}^{\infty} v_j \psi_j \otimes \psi_j$ and $\Gamma = \sum_{j=1}^{d_{\Gamma}} \zeta_j \varphi_j \otimes \varphi_j$. The next two theorems are about the convergence rates of the SDR directions corresponding to the true survival time.

Theorem 1. Under the assumptions of Lemma 4, for any $N > 0$, we have

$$\left\| \left(\hat{\Sigma}^2 + \tau I \right)^{-1} \hat{\Sigma} \hat{\Gamma} - \Sigma^{-1} \Gamma \right\|_{\text{HS}} = O_p \left(\frac{1}{\tau n^{\kappa}} \right) + \sum_{j=1}^{d_{\Gamma}} \left(\frac{\tau}{v_N^2} \left\| \Psi_N(\tilde{\varphi}_j) \right\| + \left\| \Psi_N^{\perp}(\tilde{\varphi}_j) \right\| \right), \quad (26)$$

where $\tilde{\varphi}_j = \Sigma^{-1} \varphi_j$, Ψ_N and $\Psi_N^{\perp}$ are, respectively, the projection operator and its complement,

$$\Psi_N = \sum_{j=1}^N \psi_j \otimes \psi_j, \quad \Psi_N^{\perp} = \sum_{j=N+1}^{\infty} \psi_j \otimes \psi_j.$$

If the smoothing parameter $\tau = \tau(n)$ satisfies that $\lim_{n \rightarrow \infty} \tau = 0$ and $\lim_{n \rightarrow \infty} \tau n^{\kappa} = \infty$, then

$$\left\| \left(\hat{\Sigma}^2 + \tau I \right)^{-1} \hat{\Sigma} \hat{\Gamma} - \Sigma^{-1} \Gamma \right\|_{\text{HS}} = o_p(1).$$

Remark 7. Note that in the infinite-dimensional case, Σ^{-1} is unbounded. So, for the operator $\Sigma^{-1} \Gamma$ to be well defined, we need to show that the range of Γ is indeed in the range of Σ . In fact, the operator $\Sigma^{-1} \Gamma$ is automatically well defined when the linearity condition in Lemma 1 holds. To be precise, we know that $E(\phi(\mathbf{X})|T) \in \text{span}\{\Sigma \boldsymbol{\eta}_1, \dots, \Sigma \boldsymbol{\eta}_q\}$. For any $f \in \mathcal{H}$, $\Gamma f = E(E(\phi(\mathbf{X})|T), f)_2 E(\phi(\mathbf{X})|T)$ also belongs to $\text{span}\{\Sigma \boldsymbol{\eta}_1, \dots, \Sigma \boldsymbol{\eta}_q\} \subset \text{range}(\Sigma)$. Therefore the operator $\Sigma^{-1} \Gamma$ is well defined over the whole space $\mathcal{H}$. From the decomposition (6), we have

$$\Sigma^{-1} \Gamma f = \Sigma^{-1} \left(\sum_{i=1}^{d_{\Gamma}} \zeta_i \langle \varphi_i, f \rangle_2 \varphi_i \right) = \sum_{i=1}^{d_{\Gamma}} \zeta_i \langle \varphi_i, f \rangle_2 \Sigma^{-1}(\varphi_i) = \left(\sum_{i=1}^{d_{\Gamma}} \Sigma^{-1}(\varphi_i) \otimes \varphi_i \right) f, \quad (27)$$

where $\Sigma^{-1}(\varphi_i)$ is well defined because $\varphi_i = \frac{1}{\zeta_i} \Gamma \varphi_i \in \text{range}(\Sigma)$. The expression (27) shows that $\Sigma^{-1}\Gamma$ is a Hilbert–Schmidt operator.

Theorem 2. *If $\lim_{n \rightarrow \infty} \tau = 0$, $\lim_{n \rightarrow \infty} \tau n^\kappa = \infty$, d_Γ is the rank of Γ , $\hat{\beta}_j$ is the estimate from (7), then*

$$|\hat{u}_j(\mathbf{X}) - u_j(\mathbf{X})| = |\langle \hat{\beta}_j, \boldsymbol{\phi}(\mathbf{X}) \rangle - \langle \beta_j, \boldsymbol{\phi}(\mathbf{X}) \rangle| = o_p(1), \quad j = 1, \dots, d_\Gamma.$$

Furthermore, under Assumption (A8), $\{\beta_j\}_{j=1}^{d_\Gamma}$ depend only on a finite number of eigenvectors of the covariance operator of Σ , the rate of convergence is $O(n^{-\kappa/2})$. That is

$$|\hat{u}_j(\mathbf{X}) - u_j(\mathbf{X})| = |\langle \hat{\beta}_j, \boldsymbol{\phi}(\mathbf{X}) \rangle - \langle \beta_j, \boldsymbol{\phi}(\mathbf{X}) \rangle| = O_p(n^{-\kappa/2}), \quad j = 1, \dots, d_\Gamma.$$

Theorem 2 is a corollary of Theorem 1. By a well-known result in the perturbation theory Kato (2013), the eigenspaces of $(\hat{\Sigma}^2 + \tau I)^{-1} \hat{\Sigma} \hat{\Gamma}$ converge to those of $\Sigma^{-1}\Gamma$ at the same rate. Therefore, the proof of Theorem 2 is omitted.

5 | SIMULATIONS AND REAL DATA ANALYSIS

5.1 | Simulations

In this section, we investigate the performances of RDSIR by simulation. As we have mentioned in the introduction, to the best of our knowledge there is no literature about nonlinear SDR for survival data, so we compare our method with some existing linear SDR methods to highlight that our proposed method can effectively extract nonlinearity from survival data. We compare RDSIR with double-sliced inverse regression (DSIR) proposed by Li et al. (1999) and the counting process inverse regression (IR-CP) approach (Sun et al., 2019) which are suitable for linear SDR.

Commonly used kernels are the Gaussian radial basis kernel $\mathcal{R}(\mathbf{s}, \mathbf{t}) = \exp(-\text{scale} \cdot \|\mathbf{s} - \mathbf{t}\|^2)$ and the polynomial kernel $\mathcal{R}(\mathbf{s}, \mathbf{t}) = (\text{scale} \cdot \langle \mathbf{s}, \mathbf{t} \rangle + \text{offset})^{\text{degree}}$. Duan et al. (2003) and Keerthi and Lin (2003) proposed some criteria about how to choose reproducing kernel and corresponding parameter. When there is no prior information on a dataset, the Gaussian radial basis kernel is usually chosen. In the following examples, we set training sample size $n_{tr} = 100$ and test sample size $n_{te} = 200$. The regularization parameters τ and s can be decided by the criterion (25) proposed at the end of Section 3.

To compare the estimated and true dimension reduction directions, which may be vectors of different dimensions, we adopt the *Robust Maximum Association Estimators* (RMAE) (Alfons et al., 2017) as follows. Suppose that $\mathbf{X}$ is a p -dimensional random vector and $\mathbf{Y}$ is a q -dimensional random vector, with $p \geq q$. A measure of multivariate association between $\mathbf{X}$ and $\mathbf{Y}$ can be defined by looking for linear combinations $\boldsymbol{\alpha}^T \mathbf{X}$ and $\boldsymbol{\beta}^T \mathbf{Y}$ of the original variables that has the maximal association. That is, we seek a measure

$$\rho_r(\mathbf{X}, \mathbf{Y}) = \max_{\boldsymbol{\alpha}, \boldsymbol{\beta}} r(\boldsymbol{\alpha}^T \mathbf{X}, \boldsymbol{\beta}^T \mathbf{Y}),$$

where r is a measure of association between two univariate variables. Taking the classical Pearson correlation for r results in the first canonical correlation coefficient (Johnson et al., 2002). The bivariate association measure r considered in this article is Spearman's rank correlation which is defined as

$$r(U, V) = r_P(\text{rank}(U), \text{rank}(V)),$$

where $\text{rank}(u) = F_U(u)$, with F_U the cumulative distribution function of the random variable U , stands for the population rank of u . And $r_P(X, Y) = \text{cov}(X, Y) / \sqrt{\text{var}(X)\text{var}(Y)}$ is the Pearson correlation. For practical application, an empirical version of ρ_r denoted by $\hat{\rho}_r$ is needed. Alfons et al. (2017) developed the *alternate grid* algorithm for the computation of RMAE and studied their theoretical properties for various association measures such as Pearson, Spearman, and Kendall's τ correlations. It turns out that Spearman and Kendall's τ correlations yield a maximum association estimate with good robustness properties and good efficiency, which has been implemented in R package *ccapp* (Alfons et al., 2016). In the following, we perform 100 Monte Carlo simulations for each model and report the averages and SDs of RMAE under different censoring proportions 0%, 20%, 40%, and 60%.

Model 1 (Linear). We generate survival time and censoring time from

$$\begin{aligned} T &= \Phi(\epsilon * \beta_1^T \mathbf{X}), \\ C &= \Phi(\beta_2^T \mathbf{X}) + U, \end{aligned}$$

where $\mathbf{X} \sim N(0, I_{50})$, $\beta_1 = (1, 0, -1, 0, \dots, 1, 0, -1, 0, 0, \dots, 0)^T / 5$, the first 40 components are 10 repetitions of $(1, 0, -1, 0)$ and the last 10 elements are 0 and the second direction $\beta_2 = (0, \dots, 0, 1, -1, \dots, 1, -1)^T / 5$, the first 10 elements are 0 and the last 40 components are 20 repetitions of $(1, -1)$. Φ is the cumulative distribution function of the standard normal distribution and ϵ follows a uniform distribution $U(0, 1)$. The symbol U denotes a random variable uniformly distributed on $(0, c)$, where c controls the censoring proportion. We set the reproducing kernel $\mathcal{R}(\mathbf{s}, \mathbf{t}) = \langle \mathbf{s}, \mathbf{t} \rangle_2$ and consider the first two directions. The results are given in Table 1.

Model 2 (Linear). In this example, we consider two linear SDR directions $\beta_1^T \mathbf{X}$ and $\beta_2^T \mathbf{X}$. We simulate the survival time by

$$T = -\frac{\log(U)}{\exp(\beta_1^T \mathbf{X}) + \exp(\beta_2^T \mathbf{X})},$$

where $\mathbf{X} \sim N(0, I_{50})$ and $U \sim U(0, 1)$. The coefficient vector $\beta_1 = (1, -1, \dots, 1, -1, 0, \dots, 0)^T / 5$ where the last 10 elements are 0 and the first 40 components are 20 repetitions of $(1, -1)$. The other direction $\beta_2 = (0, 0, \dots, 0, 1, -1, \dots, 1, -1)^T / 5$ where the first 10 elements are 0 and the last 40 components are 20 repetitions of $(1, -1)$. The censoring time is generated from $U(0, c)$ and c controls the censoring rate. It is obvious that this is a linear dimension reduction problem and we want to recover the space spanned by $\{\beta_1, \beta_2\}$. We set the reproducing kernel $\mathcal{R}(\mathbf{s}, \mathbf{t}) = \langle \mathbf{s}, \mathbf{t} \rangle_2$ and the results are summarized in Table 1.

It is noted that for both Model 1 and Model 2, the true SDR directions are in the 50-dimensional Euclidean space. In Table 1, we can see that DSIR and IR-CP perform well on this type of linear dimension reduction problem. We can also learn from Table 1 that RDSIR can achieve similar

TABLE 1 Mean and SD of the RMAE under different censoring proportion

			0%	20%	40%	60%
Model 1	$q = 1$	RDSIR	0.933(0.027)	0.883(0.044)	0.801(0.063)	0.728(0.091)
		DSIR	0.936(0.024)	0.895(0.036)	0.828(0.055)	0.766(0.076)
		IR-CP	0.941(0.021)	0.902(0.032)	0.836(0.049)	0.769(0.075)
	$q = 2$	RDSIR	0.939(0.024)	0.896(0.036)	0.823(0.058)	0.762(0.070)
		DSIR	0.940(0.022)	0.903(0.032)	0.843(0.053)	0.789(0.064)
		IR-CP	0.947(0.019)	0.911(0.028)	0.853(0.048)	0.804(0.060)
Model 2	$q = 1$	RDSIR	0.895(0.036)	0.886(0.035)	0.838(0.050)	0.792(0.059)
		DSIR	0.903(0.028)	0.899(0.030)	0.867(0.039)	0.834(0.045)
		IR-CP	0.908(0.024)	0.905(0.030)	0.874(0.035)	0.845(0.044)
	$q = 2$	RDSIR	0.903(0.033)	0.897(0.031)	0.854(0.046)	0.814(0.057)
		DSIR	0.910(0.027)	0.907(0.029)	0.878(0.034)	0.849(0.042)
		IR-CP	0.917(0.023)	0.910(0.024)	0.884(0.032)	0.849(0.037)

Abbreviations: DSIR, double sliced inverse regression; IR-CP, counting process inverse regression; RDSIR, reproducing kernel Hilbert space-based DSIR; RMAE, robust maximum association estimators.

performance compared to DSIR and IR-CP. In fact, the key idea of the kernel trick is to obtain the inner product in the high-dimensional feature space by computing the value of the kernel function in the original input space. When using a linear kernel, RDSIR is equivalent to DSIR. There will be a small difference between RDSIR and DSIR and IR-CP due to the accumulation of calculation errors.

Model 3 (Nonlinear). To show the performance of RDSIR when there exists a nonlinear dimension reduction subspace, we simulate the survival data by

$$T = U_1 * [\mathbf{X}^{(1)2} + \dots + \mathbf{X}^{(50)2}],$$

$$C = U_2 * [\sin(\mathbf{X}^{(1)}) + \dots + \sin(\mathbf{X}^{(50)})]^2,$$

where $\mathbf{X} \sim N(0, I_p)$, $p = 50, 60, 70$, $X^{(i)}$ is the i th element of $\mathbf{X}$. The random error $U_1 \sim U(0, 1)$, $U_2 \sim U(0, c)$ and c controls the censoring proportion. Let the reproducing kernel be $\mathcal{R}(s, t) = (\langle s, t \rangle + 1)^2$, see Table 2 for the simulation results.

Model 4 (Nonlinear). We consider another nonlinear dimension reduction example which exists two directions. The survival data is simulated by

$$T = -\frac{\log(U)}{\sin^2(\mathbf{X}^{(1)}/2) + \sin^2(\mathbf{X}^{(2)}/2) + \dots + \sin^2(\mathbf{X}^{(50)}/2)} + \exp\left\{-\frac{1}{2}\mathbf{X}^T\mathbf{X}\right\},$$

where the random variable $U \sim U(0, 1)$ and censoring time $C \sim U(0, c)$ with c controlling the censoring rate, and $\mathbf{X} \sim N(0, I_p)$, $p = 50, 60, 70$. We choose the Gaussian radial basis kernel $\mathcal{R}(s, t) = \exp(-\text{scale} \cdot \|s - t\|^2)$ and set $\text{scale} = 0.1$. The results are displayed in Table 3. What we need to

TABLE 2 The RMAE mean and SD under different censoring proportion.

			0%	20%	40%	60%
Model 3 ($n = 100, p = 50$)	$q = 1$	RDSIR	0.966(0.033)	0.928(0.107)	0.898(0.186)	0.866(0.213)
		DSIR	0.065(0.051)	0.055(0.038)	0.068(0.048)	0.062(0.043)
		IR-CP	0.176(0.043)	0.150(0.047)	0.097(0.042)	0.066(0.041)
	$q = 2$	RDSIR	0.972(0.015)	0.948(0.075)	0.934(0.120)	0.929(0.099)
		DISR	0.103(0.050)	0.086(0.041)	0.106(0.044)	0.100(0.043)
		IR-CP	0.186(0.053)	0.165(0.057)	0.128(0.056)	0.116(0.053)
Model 3 ($n = 100, p = 60$)	$q = 1$	RDSIR	0.873(0.024)	0.837(0.115)	0.814(0.128)	0.787(0.168)
		DSIR	0.055(0.042)	0.051(0.046)	0.058(0.050)	0.051(0.045)
		IR-CP	0.124(0.037)	0.105(0.045)	0.098(0.048)	0.081(0.045)
	$q = 2$	RDSIR	0.877(0.021)	0.859(0.057)	0.848(0.069)	0.840(0.079)
		DSIR	0.097(0.044)	0.089(0.048)	0.094(0.052)	0.093(0.048)
		IR-CP	0.135(0.041)	0.119(0.046)	0.108(0.050)	0.099(0.045)
Model 3 ($n = 100, p = 70$)	$q = 1$	RDSIR	0.783(0.092)	0.757(0.122)	0.724(0.147)	0.695(0.169)
		DSIR	0.059(0.043)	0.049(0.040)	0.061(0.045)	0.062(0.046)
		IR-CP	0.105(0.044)	0.096(0.045)	0.088(0.050)	0.081(0.049)
	$q = 2$	RDSIR	0.799(0.042)	0.790(0.050)	0.768(0.083)	0.757(0.092)
		DSIR	0.103(0.042)	0.090(0.040)	0.099(0.043)	0.093(0.044)
		IR-CP	0.113(0.043)	0.107(0.044)	0.098(0.040)	0.089(0.045)

Abbreviations: DSIR, double sliced inverse regression; IR-CP, counting process inverse regression; RDSIR, reproducing kernel Hilbert space-based DSIR; RMAE, robust maximum association estimators.

pay attention to is that both nonlinear SDR directions $u_1(\mathbf{X}) = \sin^2(\mathbf{X}^{(1)}/2) + \sin^2(\mathbf{X}^{(2)}/2) + \dots + \sin^2(\mathbf{X}^{(50)}/2)$ and $u_2(\mathbf{X}) = \mathbf{X}^T \mathbf{X}$ are artificially assumed in the model setting stage. We can also regard these two directions as one direction since they can be absorbed in the unknown function g . Thus, we may find that the first estimated direction contains most of the information and there is no significant improvement in the results after adding the second direction. On the other hand, it is also possible that the first two directions are not enough to characterize the relationship between T and $\mathbf{X}$. So, the key problem is to choose a suitable dimension q which has been discussed in Section 3.

By taking these examples into account, we can conclude that the larger the censoring rate and the number of regressors, the harder the problem and one might expect the performance of our method to deteriorate. From Tables 1 to 3, we can see that DSIR and IR-CP fail to extract nonlinear dimension reduction directions, but RDSIR proposed in this paper performs well in both linear and nonlinear situations. Since our proposed method is an extension of DSIR, we show in Table 4 the averaged computation time (in seconds) of the two methods based on 100 repetitions for Model 4. We can see that the computation efficiency does not severely reduce due to the use of the kernel and the selection of regularization parameters.

TABLE 3 Mean and SD of the RMAE under different censoring proportion.

			0%	20%	40%	60%
Model 4 ($n = 100, p = 50$)	$q = 1$	RDSIR	0.895(0.125)	0.884(0.176)	0.873(0.178)	0.851(0.187)
		DSIR	0.101(0.040)	0.102(0.043)	0.103(0.041)	0.105(0.044)
		IR-CP	0.093(0.047)	0.081(0.039)	0.085(0.040)	0.075(0.044)
	$q = 2$	RDSIR	0.934(0.057)	0.932(0.092)	0.930(0.067)	0.901(0.134)
		DISR	0.136(0.039)	0.147(0.043)	0.141(0.043)	0.140(0.046)
		IR-CP	0.117(0.041)	0.112(0.043)	0.103(0.044)	0.096(0.054)
model 4 ($n = 100, p = 60$)	$q = 1$	RDSIR	0.826(0.053)	0.800(0.084)	0.760(0.142)	0.719(0.184)
		DSIR	0.092(0.044)	0.106(0.060)	0.082(0.039)	0.108(0.057)
		IR-CP	0.100(0.038)	0.102(0.046)	0.099(0.041)	0.093(0.044)
	$q = 2$	RDSIR	0.855(0.029)	0.839(0.042)	0.809(0.105)	0.806(0.064)
		DSIR	0.141(0.050)	0.138(0.045)	0.130(0.044)	0.135(0.067)
		IR-CP	0.112(0.040)	0.105(0.043)	0.101(0.042)	0.095(0.043)
Model 4 ($n = 100, p = 70$)	$q = 1$	RDSIR	0.758(0.050)	0.721(0.078)	0.708(0.118)	0.631(0.224)
		DSIR	0.089(0.031)	0.110(0.048)	0.100(0.028)	0.095(0.024)
		IR-CP	0.092(0.030)	0.087(0.033)	0.083(0.031)	0.075(0.036)
	$q = 2$	RDSIR	0.766(0.049)	0.748(0.049)	0.737(0.075)	0.702(0.142)
		DSIR	0.144(0.035)	0.150(0.040)	0.140(0.039)	0.127(0.040)
		IR-CP	0.107(0.031)	0.103(0.033)	0.098(0.031)	0.095(0.034)

Abbreviations: DSIR, double sliced inverse regression; IR-CP, counting process inverse regression; RDSIR, reproducing kernel Hilbert space-based DSIR; RMAE, robust maximum association estimators.

5.2 | NCCTG lung cancer data

Consider the North Central Cancer Treatment Group (NCCTG) lung cancer data in Loprinzi et al. (1994). NCCTG lung cancer dataset recorded the survival days of patients with advanced lung cancer, together with assessments of the patients' performance status measured either by the physician and by the patients themselves. There were total 228 patients including 63 patients whose data were right censored. Ten variates are included in our study, specifically, survival time in days (time: T), censoring status 1 = censored and 2 = dead (status: δ), institution code (inst: X_1), age in years (age: X_2), male = 1 and female = 2 (sex: X_3), ECOG performance score (0 = good to 5 = dead) (ph.ecgo: X_4), Karnofsky performance score (bad=0 to good = 100) rated by physician (ph.karno: X_5), Karnofsky performance score as rated by patient (pat.karno: X_6), calories consumed at meals (meal.cal: X_7), weight loss in last 6 months (wt.loss: X_8).

We fit the Cox proportional hazards (PH) model using these covariates and all the interaction terms. The Harrell's C-index is 0.725 which means that this model fits the data well. We also plot the Kaplan–Meier estimate of the survival curves for the three risk groups determined by the risk scores (divided by the 33% and 66% quantiles) estimated from the Cox PH model with all covariates and interaction terms. The result is displayed in Figure 1. From this figure, it can be

TABLE 4 The averaged computation time (in s) based on 100 repetitions for Model 4.

			0%	20%	40%	60%
Model 4 ($n = 100, p = 50$)	$q = 1$	RDSIR	0.995	1.004	1.227	1.351
		DSIR	0.732	0.893	1.003	1.115
	$q = 2$	RDSIR	1.076	1.152	1.274	1.444
		DISR	0.935	0.981	1.069	1.158
model 4 ($n = 100, p = 60$)	$q = 1$	RDSIR	1.124	1.293	1.496	1.587
		DSIR	1.011	1.105	1.298	1.374
	$q = 2$	RDSIR	1.182	1.364	1.571	1.683
		DSIR	1.075	1.184	1.364	1.507
Model 4 ($n = 100, p = 70$)	$q = 1$	RDSIR	1.354	1.521	1.708	1.805
		DSIR	1.289	1.443	1.600	1.695
	$q = 2$	RDSIR	1.466	1.548	1.837	2.002
		DSIR	1.340	1.490	1.701	1.867

Abbreviations: DSIR, double sliced inverse regression; IR-CP, counting process inverse regression; RDSIR, reproducing kernel Hilbert space-based DSIR; RMAE, robust maximum association estimators.

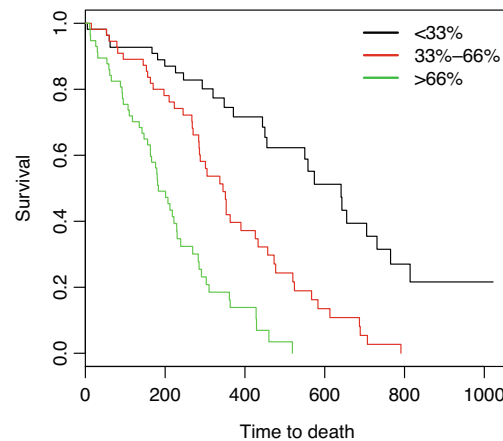


FIGURE 1 The Kaplan-Meier estimate of survival curves for the three risk groups determined by the risk scores (divided by the 33% and 66% quantiles) estimated from the Cox proportional hazards model with all covariates

seen that the three risk curves are well separated, which also suggests that the Cox PH model fits the data reasonably well.

For applying the proposed method to the data, we choose the Gaussian kernel and set the regularity parameter $\tau = 1$. We denote $\hat{v}_1, \hat{v}_2$ as the first two directions or components found by DSIR or RDSIR. The Cox PH model we fit is $\lambda(t) = \lambda_0(t) \exp\{\theta_1 v_1 + \theta_2 v_2 + \theta_3 v_1 v_2\}$, where $\lambda(t)$ is hazard function and $\lambda_0(t)$ is baseline hazard function. The results are displayed in Figure 2. From

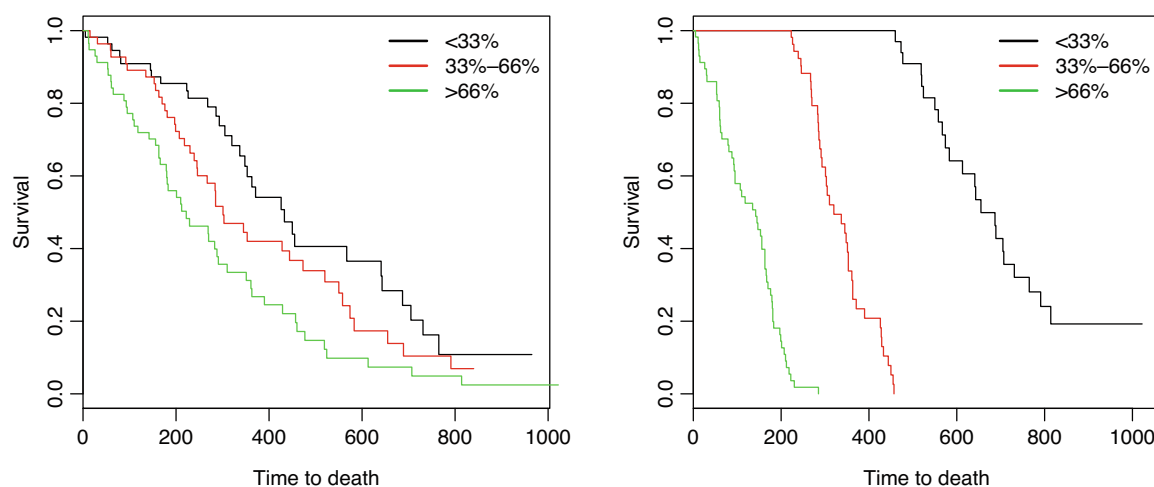


FIGURE 2 The Kaplan–Meier estimate of survival curves for the three risk groups determined by the risk scores (divided by the 33% and 66% quantiles) estimated from the Cox proportional hazards model with double sliced inverse regression directions (DSIR; left) and reproducing kernel Hilbert space-based DSIR directions (right).

this figure, we can see that the three risk group curves corresponding to RDSIR directions show a better separation than the ones corresponding to DSIR directions, which indicates that RDSIR outperforms DSIR.

6 | CONCLUSION

In this paper, we give a new nonlinear SDR method for survival data and demonstrate its superiority over DSIR via numerical studies. Our method is based on DSIR and RKHS theory. We adopt the double slicing procedure to estimate weight function which is an important step for modeling survival data. The estimation of weight function requires nonparametric smoothing. We tackle it by using kernel smoothing in this paper for the convenience of theoretical development. However, other nonparametric smoothing methods such as spline (De Boor et al., 1978; Wahba, 1990) and wavelets (Ogden, 2012; Vidakovic, 2009) can also be used. Another idea for handling survival data is to impute the censored T observations first and then apply the SDR methods directly to the imputed data.

The choice of reproducing kernel is also worth taking into account. In general settings, we are not limited to using a single kernel. Alternatively, we can learn from the idea of multiple kernel learning (Gönen & Alpaydin, 2011; Sonnenburg et al., 2006). As an example, consider a linear combination of m kernels, that is, $K' = \sum_{i=1}^m K_i$ and this new function is still a kernel due to the properties of RKHS. Relevant theories and experiments on this topic deserve further study.

Other inverse regression methods such as SAVE (Cook & Weisberg, 1991) can also be generalized to nonlinear SDR for survival data in a similar way. Other asymptotic properties such as asymptotic normality need also further investigation. The case where p diverges with n at some rate will be considered in our future research.

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CONFLICT OF INTEREST

None of us, Wenquan Cui, Jianjun Xu and Yuehua Wu, declares financial conflicts.

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